Lösung: gemischte Integrationsregeln

a)

$$\int \underbrace{\sin(x)}_{u(x)} \cdot \underbrace{\cos(x)}_{v'(x)}\, dx = \underbrace{\sin(x)}_{u(x)} \cdot \underbrace{\sin(x)}_{v(x)} - \int \underbrace{\cos(x)}_{u'(x)} \cdot \underbrace{\sin(x)}_{v(x)}\, dx + C$$

$$\underbrace{\int \sin(x)\cdot\cos(x)\, dx}_{I} = \sin^2(x) - I + C$$

$$2I = \sin^2(x) + C$$

$$I = \int \sin(x)\cdot\cos(x)\, dx = \frac{\sin^2(x)}{2} + \tilde{C}$$

b)

$$\int \cos^2(x)\, dx = \int \underbrace{\cos(x)}_{u(x)} \cdot \underbrace{\cos(x)}_{v'(x)}\, dx$$

$$\underbrace{\int \cos^2(x)\, dx}_{I} = \underbrace{\cos(x)}_{u(x)} \cdot \underbrace{\sin(x)}_{v(x)} - \int \underbrace{(-\sin(x))}_{u'(x)} \cdot \underbrace{\sin(x)}_{v(x)}\, dx$$

$$= \cos(x)\sin(x) + \int \sin^2(x)\, dx$$

mit $\cos^2(x) + \sin^2(x) = 1$ folgt

$$= \cos(x)\sin(x) + \int (1 - \cos^2(x))\, dx$$

$$= \cos(x)\sin(x) + \underbrace{\int dx}_{= x + C} - \underbrace{\int \cos^2(x)\, dx}_{= I}$$

$$\Rightarrow \quad I = \cos(x)\sin(x) + x + C - I$$

$$2I = \cos(x)\sin(x) + x + C$$

$$I = \frac{\cos(x)\sin(x) + x}{2} + \frac{C}{2}$$

$$I = \int \cos^2(x)\,dx = \frac{\cos(x)\sin(x) + x}{2} + \tilde{C}$$

c) $v'(x)$, $u(x)$; $v(x)$, $u(x)$; $v(x)$, $u'(x)$

$$\int x \cdot \ln(x)\,dx = \frac{x^2}{2}\cdot \ln(x) - \int \frac{x^2}{2}\cdot\frac{1}{x}\,dx$$

$$= \frac{x^2}{2}\ln(x) - \frac{1}{2}\int x\,dx$$

$$= \frac{x^2}{2}\ln(x) - \frac{x^2}{4} + C$$

$$\int x \cdot \ln(x)\,dx = \frac{x^2}{4}(2\ln(x) - 1) + C$$

d) $\int \frac{x^2}{\sqrt{1-x^3}}\,dx$ mit $z = 1 - x^3$ $\left(\frac{dz}{dx} = -3x^2\right)$

$$\ldots = \int \frac{x^2}{\sqrt{z}}\frac{dz}{(-3x^2)} = \frac{-1}{3}\int \frac{dz}{\sqrt{z}} = \frac{-1}{3}\int z^{-1/2}\,dz$$

$$= \frac{-1}{3}\,\frac{1}{1/2}\,z^{1/2} + C = -\frac{2}{3}\sqrt{z} + C$$

$$\int \frac{x^2}{\sqrt{1-x^3}}\,dx = -\frac{2}{3}\sqrt{1-x^3} + C$$

e) $\int \cos^3(x)\cdot\sin(x)\,dx$ mit $z = \cos(x)$ $\left(\frac{dz}{dx} = -\sin(x)\right)$

$$\ldots = \int z^3 \cdot \sin(x)\,\frac{dz}{(-\sin(x))} = -\int z^3\,dz = -\frac{z^4}{4} + C$$

$$= -\frac{\cos^4(x)}{4} + C$$

f)

$$\int x^2 \cdot e^{x^3-2}\,dx \qquad \text{mit} \quad z = x^3-2 \quad \left(\frac{dz}{dx} = 3x^2\right)$$

$$\ldots = \int x^2 \cdot e^z \frac{dz}{3x^2} = \frac{1}{3}\int e^z dz = \frac{e^z}{3} + C$$

$$= \underline{\underline{\frac{e^{(x^3-2)}}{3} + C}}$$

g)

$$\int \overset{u(x)}{\cos(x)} \cdot \overset{v'(x)}{e^x}\,dx = \overset{u(x)}{\cos(x)} \cdot \overset{v(x)}{e^x} + C - \int \overset{u'(x)}{(-\sin(x))} \cdot \overset{v(x)}{e^x}\,dx$$

$$I = \cos(x)\cdot e^x + C + \int \overset{u(x)}{\sin(x)} \cdot \overset{v'(x)}{e^x}\,dx$$

$$= \cos(x)e^x + C + \left(\overset{u(x)}{\sin(x)}\overset{v(x)}{e^x} - \underbrace{\int \overset{u'(x)}{\cos(x)}\overset{v(x)}{e^x}dx}_{=I}\right)$$

$$\Rightarrow \quad I = e^x(\cos(x)+\sin(x)) + C - I$$

$$2I = e^x(\cos(x)+\sin(x)) + C$$

$$\Rightarrow \quad \underline{\underline{I = \int \cos(x)\cdot e^x\,dx = \frac{e^x}{2}(\cos(x)+\sin(x)) + \tilde{C}}}$$

h)

$$\int \cos(x)\cdot e^{\sin(x)}\,dx \qquad z = \sin(x) \quad \left(\frac{dz}{dx} = \cos(x)\right)$$

$$\ldots = \int \cos(x)\cdot e^z \frac{dz}{\cos(x)} = \int e^z dz = e^z + C$$

$$\underline{\underline{\int \cos(x)\cdot e^{\sin(x)}\,dx = e^{\sin(x)} + C}}$$

(i)

$(\arcsin(x))' = \frac{1}{\sqrt{1-x^2}}$

$$\int \arcsin(x)\,dx = \int \overset{u'(x)}{1} \cdot \overset{v(x)}{\arcsin(x)}\,dx$$

$$= \overset{u(x)}{x} \cdot \overset{v(x)}{\arcsin(x)} - \int \overset{u(x)}{x} \cdot \overset{v'(x)}{\frac{1}{\sqrt{1-x^2}}}\,dx$$

$$= x \cdot \arcsin(x) - \int \frac{x}{\sqrt{1-x^2}}\,dx$$

Substitution $z = 1-x^2$ $\left(\frac{dz}{dx} = -2x\right)$

$$= x \cdot \arcsin(x) - \int \frac{x}{\sqrt{z}} \frac{dz}{(-2x)} = x \cdot \arcsin(x) + \frac{1}{2}\int z^{-1/2}\,dz$$

$$= x \cdot \arcsin(x) + \frac{1}{2}\,\frac{1}{1/2}\,z^{1/2} + C$$

$$\Rightarrow \quad \underline{\int \arcsin(x) = x \cdot \arcsin(x) + \sqrt{1-x^2} + C}$$